Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/22/2019
PI Name:	Matin, A.C. Ph.D.		
Project Title:	Spaceflight Effects on Bacterial Antibiotic Resistance and Its Genetic Basis “AntimicrobialSat”		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	94305-5101	**Congressional District:**	14
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 SALMON--Astrobiology & FSB NNH08ZDA009O-SCMAFSB
Start Date:	07/01/2010	**End Date:**	12/31/2018
No. of Post Docs:	4	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	Small Satellites		
Flight Assignment:	EcAMSat NOTE: End date change to 12/31/2018 per F. Hernandez/ARC (Ed., 6/26/18) NOTE: End date changed to 6/30/2018 with new grant number 80NSSC17M0006 in August 2017 (Ed., 5/22/18) NOTE: End date changed to 04/30/2017 per F. Hernandez/ARC and NSSC information (Ed., 4/12/17) NOTE: End date changed to 3/31/2017 per PI and NSSC information (Ed., 6/13/16) NOTE: End date changed to 3/31/2016 per NSSC information (Ed., 1/20/16) NOTE: End date changed to 11/30/2015 per NSSC information and A. Chu/ARC (Ed., 12/16/14) NOTE: New end date is 11/30/2014 per NSSC information (Ed., 4/4/14) NOTE: New end date is 4/30/2014 per A. Chu/ARC and NSSC (Ed., 2/21/2013)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			

Grant/Contract No.:	80NSSC17M0006 ; NNX10AM90A
Performance Goal No.:	
Performance Goal Text:	
Task Description:	This project has accomplished one of its two milestones, and the updated summary provided now reflects not only the conceptual genesis of the project, but also as it was refined as the project progressed. While, of course, the original essence remains, what is presented here is the summit of the first milestone, and precedes previous reports. This milestone is concerned with Earth studies to develop and rigorously test the technology for determining the sensitivity of uropathogenic Escherichia coli (UPEC) to the antibiotic gentamicin (Gm) during spaceflight, and to elucidate the genetic basis of whatever change is seen. The second milestone is experimentation during the spaceflight itself, which was launched in late 2017. A paper entitled: “Payload hardware and experimental protocol development to enable future testing of the effect of space microgravity on the resistance to gentamicin of uropathogenic Escherichia coli and its (sigma)s-deficient mutant,” was published online in May 2017: Life Sciences in Space Research 15 (2017) 1–10 2214-5524/ © 2017 The Committee on Space Research (COSPAR). Published by Elsevier Ltd.; [http://dx.doi.org/] As has been pointed out before, this project represents highly collaborative and collegial collaboration between Stanford and NASA Ames Research Center at Moffett Field. The published paper reflects this not only by the joint authorship of it by both Stanford and Moffett Field scientists, but also the benefits that accrue from collaboration between biological expertise (Stanford) and the flight technological expertise (Moffett Field). The list of authors is as follows: AC Matin, J-H Wang, Mimi Keyhan, Rachna Singh, and Michael Benoit of the Department of Microbiology & Immunology at Stanford School of Medicine; and Macarena P. Parra, Michael R. Padgen, Antonio J. Ricco, Matthew Chin, Charlie R. Friedericks, Tori N. Chinn, Aaron Cohen, Michael B. Henschke, Timothy V. Snyder, Matthew P. Lera, Shannon S. Ross, Christina M. Mayberry, Sungshin Choi, Diana T. Wu, Ming X. Tan, Travis D. Boone, Christopher C. Beasley, and Stevan M. Spremo, of NASA Ames Research Center, Moffett Field, CA 94035, USA. The abstract of this paper, reproduced here (in quotes), succinctly describes the evolved conceptual background, and demonstrates in detail the development of the EcAMSat platform, and its thorough testing using spaceflight-like commands. “Human immune response is compromised and bacteria can become more antibiotic resistant in space microgravity (MG). We report that under low-shear modeled microgravity (LSMMG), stationary-phase uropathogenic Escherichia coli (UPEC) become more resistant to gentamicin (Gm), and that this increase is dependent on the presence of s (a transcription regulator encoded by the rpoS gene). UPEC causes urinary tract infections (UTIs), reported to afflict astronauts; Gm is a standard treatment, so these findings could impact astronaut health. Because LSMMG findings can differ from MG, we report preparations to examine UPEC’s Gm sensitivity during spaceflight using the E. coli Anti-Microbial Satellite (EcAMSat) on a free-flying “nanosatellite” in low Earth orbit. Within EcAMSat’s payload, a 48-microwell fluidic card contains and supports study of bacterial cultures at constant temperature; optical absorbance changes in cell suspensions were made at three wavelengths for each microwell and a fluid-delivery system provided growth medium and predefined Gm concentrations. Performance characterization is reported here for spaceflight prototypes of this payload system. Using conventional microtiter plates, we show that Alamar Blue (AB) absorbance changes can assess the Gm effect on E. coli viability, permitting telemetric transfer of the spaceflight data to Earth. Laboratory results using payload prototypes are consistent with wellplate and flask findings of differential sensitivity of UPEC and its delta rpoS strain to Gm. if (sigma)s plays the same role in space MG as in LSMMG and Earth gravity, countermeasures discovered in recent Earth studies (aimed at weakening the UPEC antioxidant defense) to control UPEC infections would prove useful also in space flights. Further, EcAMSat results should clarify inconsistencies from previous space experiments on bacterial antibiotic sensitivity and other issues.” The manuscript may be consulted for further details and all the relevant references; see Bibliography section in 2017 report. In addition, the work has brought to light issues, such as the stress exerted by the flight equipment on bacteria, which we are now in the process of elucidating and ameliorating for the development of better future platforms for spaceflight experimentation on bacterial biology, resistance, and virulence.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Antibiotic resistance in bacteria is becoming an extremely serious problem on Earth that impacts space travelers harder given the deleterious effect of microgravity on human immune response. The research has confirmed the hypothesis that underpinned it, and has shown the way to enhance effectiveness of antibiotics in general. Subsequent work, which is in progress, has identified compounds whose co-administration with antibiotics will enhance their curative power; these can also permit lowering the needed amount of antibiotics for treating infections, thereby minimizing their side effects. This promises to improve life in space as well as on Earth.
Task Progress:	The spaceflight took place onboard a free-flying “nanosatellite,” which, for the first time utilized a 6-U configuration (a “U” being a 10-cm cube). The experiments were conducted on the EcAMSat hardware platform that we developed and described previously in the task book reports. Cell viability in this platform was measured by changes in the absorption of the dye Alamar Blue (AB); these were telemetered to Earth and recorded by daily communications with the spacecraft at Santa Clara University. Parallel ground controls were run. The flight performed very smoothly and all the needed data were successfully obtained. The major conclusion from the results is that the gene rpoS continues to play a critical role in making urinary pathogenic Escherichia coli (UPEC) highly resistant to the antibiotic gentamicin (Gm). This antibiotic is used to treat urinary tract infections often reported in astronauts. Therefore, the potential small molecule inhibitors that we have identified (using bioinformatics approaches) of the protein encoded by rpoS and the antioxidant proteins it controls hold the promise to promote astronaut health in spaceflights. That these molecules are valuable also in meeting the rampant bacterial antibiotic resistance on Earth has been pointed out before. A second important finding is that the high-aspect-ratio vessels (HARVs) used to simulate microgravity on Earth did indeed provide correct information of UPEC biology in this respect because it is with the use of this system that we had first discovered the rpoS role in Gm resistance. The cumulative work on this grant earned the entire EcAMSat team individual NASA awards.

	The publication concerning this work will be submitted within the coming weeks.
Bibliography Type:	Description: (Last Updated: 03/27/2020)
Articles in Peer-reviewed Journals	Padgen MR, Lera MP, Parra MP, Ricco AJ, Chin M, Chinn TN, Cohen A, Friedericks CR, Henschke MB, Snyder TV, Spremo SM, Wang J-H, Matin AC. "EcAMSat spaceflight measurements of the role of σs in antibiotic resistance of stationary phase Escherichia coli in microgravity." Life Sci Space Res. 2020 Feb;24:18-24. Epub 2019 Oct 31. https://doi.org/10.1016/j.lssr.2019.10.007 ; PubMed PMID: 31987476 , Feb-2020